

Practical 2 guidelines

Print this out and bring it to the practical. You need to complete this during the practical session. After completing the form, log on to ClickUP and fill in the answers online (Practical 2 Online Evaluation).

NB: This form will not count any marks, only the answers you submit to ClickUP will count marks. This file is merely here for your convenience - since I give many additional hints and help in this file.

NB: Use the decimal point for number separation. Do not use the comma. eg. 2.1 instead of 2,1. Marks will be lost if the comma is used. Also, do not supply units!
Supply answers correct to at least 2 decimal places.

On your computer Desktop is a folder labeled “NI”. In this folder is the file “NI-ELVIS Traditional”. Double click on this icon. This will start the software

Question 1

For question 1, you will measure the accuracy of the Variable Power Supply (VPS). You will do this by connecting the VPS directly to the digital mutimeter (DMM).

Instructions (Part 1):

Construct the circuit as shown in **Figure 1** by performing the following steps:

- Connect the VPS + to DMM Voltage High.
- Connect the ground to DMM Voltage Low.
- Set the VPS+ output to the following output voltages: 1, 2, 3, 4, 5 Volts
- Measure the output with the DMM.
- Complete the table below figure 1.

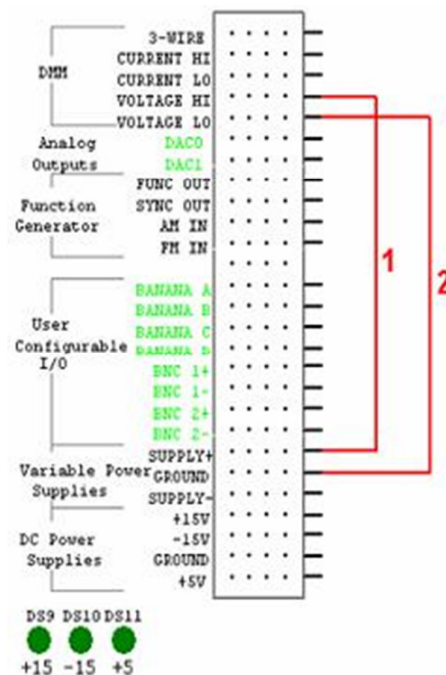


Figure 1

Supply- Voltage (in Volts)	Measured Voltage (in Volt)
1	
2	
3	
4	
5	

Instructions (Part 2):

Construct the circuit as shown in **Figure 2** by performing the following steps:

- Connect the VPS - to DMM Voltage High.
- Connect the ground to DMM Voltage Low.
- Set the **VPS**- output to the following output voltages: 1, 2, 3, 4, 5 Volts.

HINT: Remember, you are now working with negative voltages, so therefore you must use Supply – and not Supply + as in part 1.

- Measure the output with the DMM.
- Complete the table below figure 2.

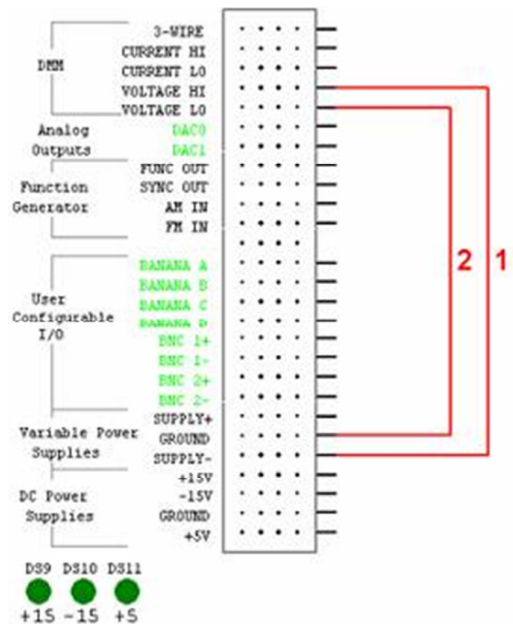


Figure 2

Supply- Voltage (in Volts)	Measured Voltage (in Volt)
-1	
-2	
-3	
-4	
-5	

Question 2

For question 2, you will measure resistance using the digital multimeter.

Instructions:

Construct the circuit as shown in **Figure 3** by performing the following steps:

- Connect the DMM Current High terminal to one side of the resistor.
- Connect the DMM Current Low terminal to the other side of the resistor
- Measure the resistance of all 3 resistors (the figure shows how to measure the resistance of the 10 Ω resistor).

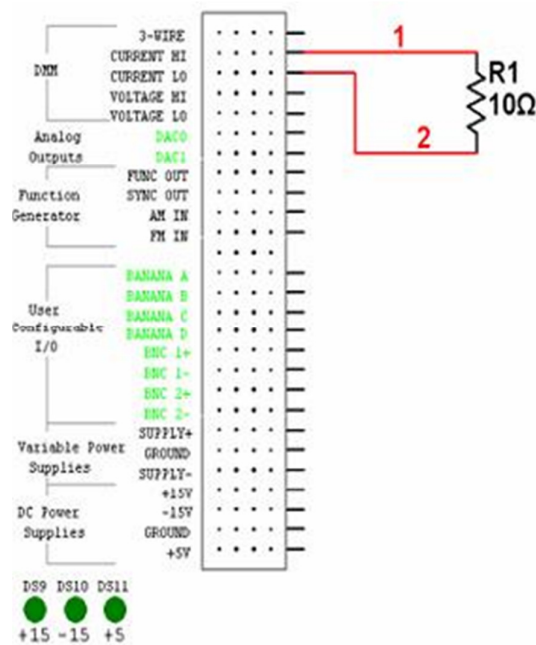


Figure 3

	Specified resistance	Measured resistance
R1	10 Ω	
R2	100 Ω	
R3	470 Ω	

Question 3

For question 3, you will measure the accuracy of the function generator. Remember that since the function generator produces a time-varying signal (sine wave/square wave/saw-tooth wave), it is necessary to do the measurements with the **Oscilloscope**.

Instructions (Part 1):

Construct the circuit as shown in **Figure 4** by performing the following steps:

- Connect the Function Generator FUNC OUT terminal to the Oscilloscope CH A+ terminal.
- Connect Ground to the Oscilloscope CH A- terminal.
- Set the amplitude of the signal to 1 V in the Function Generator window.
- Measure the output frequency for the **square** waveforms at these frequencies:
10 Hz; 100 Hz; 1 kHz; 5 kHz; 10 kHz
- Complete the table below figure 4.

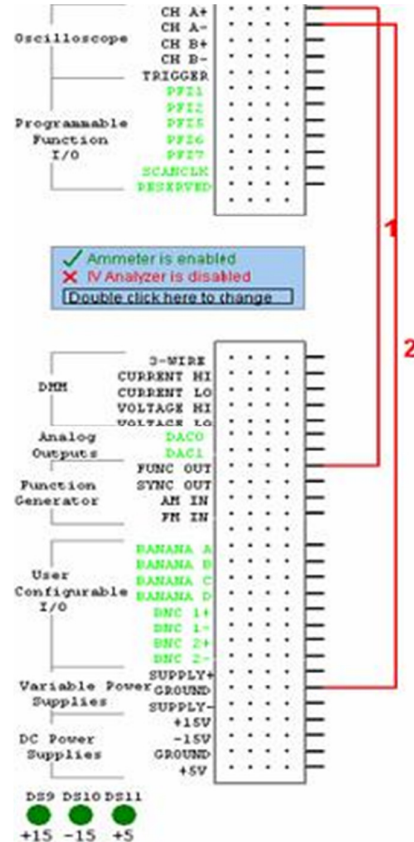


Figure 4

Hint: The actual frequency of the waveform is displayed on the oscilloscope and not in the function generator window! Make sure you take your measurements from the oscilloscope window.

Function generator frequency (Hertz)	Measured frequency (Hertz)
10	
100	
1000	
5000	
10000	

Instructions (Part 2):

Use the same circuit as in **figure 4**

- Set the frequency of the signal to 1 kHz in the Function Generator window.
- Measure the output voltage for the **sine** waveform at these amplitudes:
0.5V; 1V; 1.5V; 2V; 2.5V
- Complete the table below.

Hint: The actual amplitude of the waveform is displayed on the oscilloscope and not in the function generator window! Make sure you take your measurements from the oscilloscope window.

Also remember that the Oscilloscope displays **peak-to-peak Amplitude**, but you are asked to supply **peak amplitude**!!

Input sine wave amplitude (Volt)	Measured peak amplitude (V)
0.5	
1	
1.5	
2	
2.5	

Question 4

For question 4, you will build a simple voltage divider circuit.

Instructions (Part 1):

- Construct the circuit as shown in Figure 5.
- **HINT:** Connect the DMM Voltage High terminal to the one side of resistor R1.
- **HINT:** Connect the DMM Voltage Low terminal to the other side of resistor R1.
(I.e. we are now measuring the voltage drop across resistor R1.)
- Measure the **voltage drop** across resistor R1 for the following VPS voltages:
1V; 3V; 5V; 7 V
- Complete the table below figure 5.

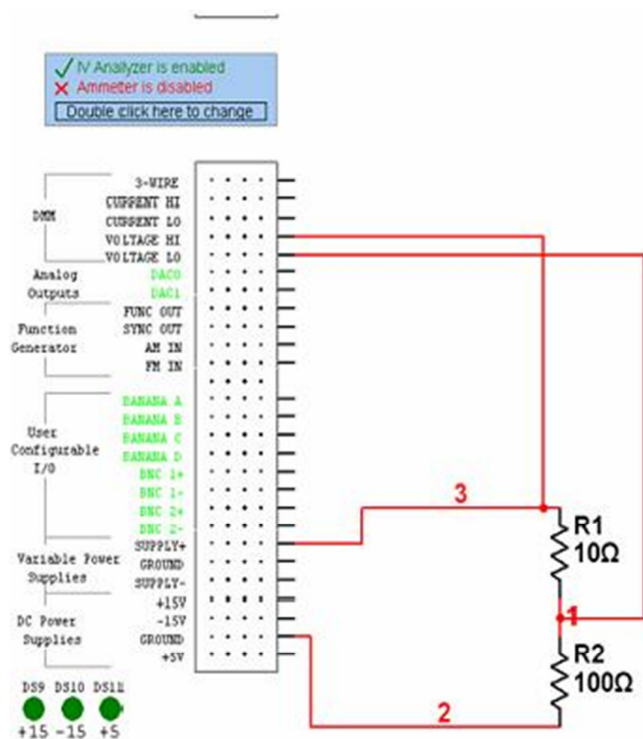


Figure 5

Supply voltage (Volt)	Voltage drop across R1 (V)
1	
3	
5	
7	

Instructions (Part 2):

- Construct the circuit as shown in Figure 6.
- Connect the DMM Current High terminal to the Variable Power Supply.
- Connect the DMM Current Low terminal to resistor R1.
- Measure the **current** through resistor R1 for the following VPS voltages: 1V; 3V; 5V; 7 V
- Complete the table below figure 6.

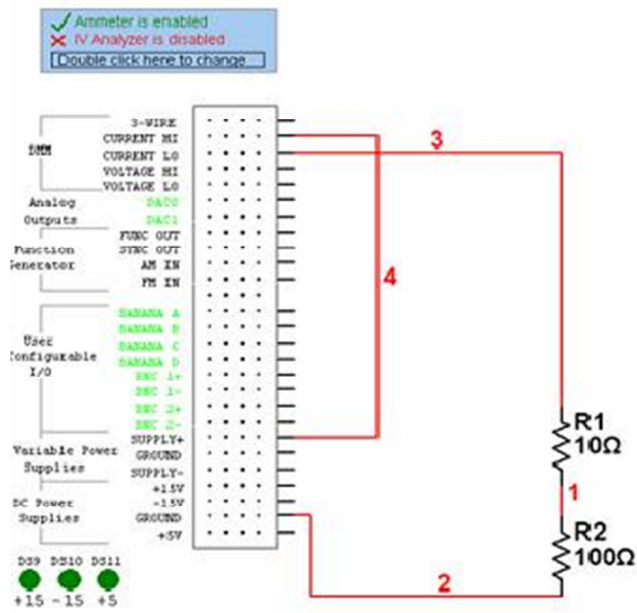


Figure 6

Supply voltage (Volt)	Current through circuit (Ampere)
1	
3	
5	
7	

Question 5

For question 5, you will build a complex voltage divider circuit.

Instructions:

- Construct the circuit as shown in Figure 7.
- Connect the DMM Voltage Meter across the parallel combination of R1 and R2.
- Measure the voltage drop across resistor R2 for the following VPS voltages:
1V; 3V; 5V; 7 V
- Complete the table below figure 7.

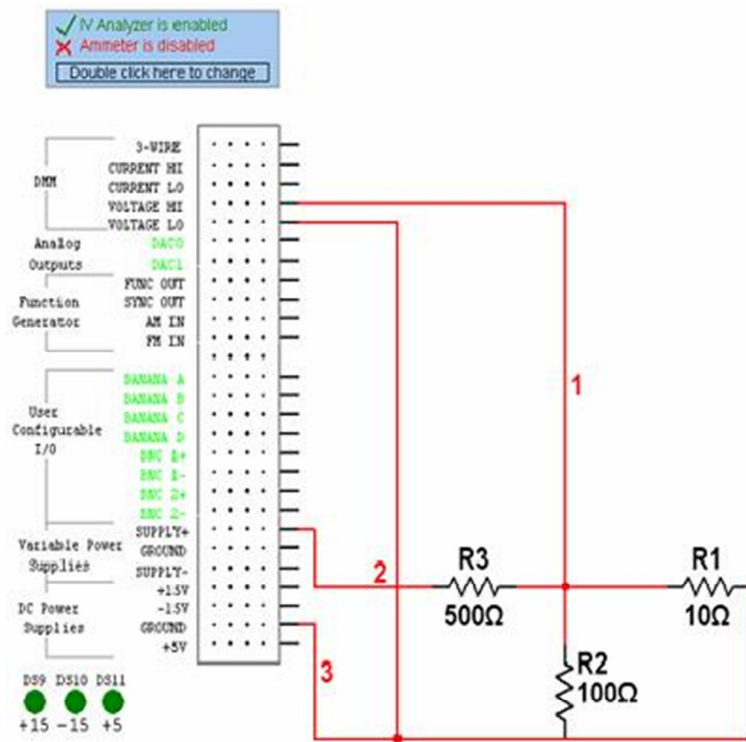


Figure 7

Supply voltage (Volt)	Voltage across parallel branch (Volt)
1	
3	
5	
7	

Question 6

For question 6, you will build a slightly more difficult voltage divider circuit.

Instructions (Part 1):

- Construct the circuit as shown in Figure 8.
- Connect the DMM Voltage Meter across resistor R3.
- Measure the **voltage** across resistor R3 for the following VPS voltages: 1V; 3V; 5V; 7 V
- Complete the table below figure 8.

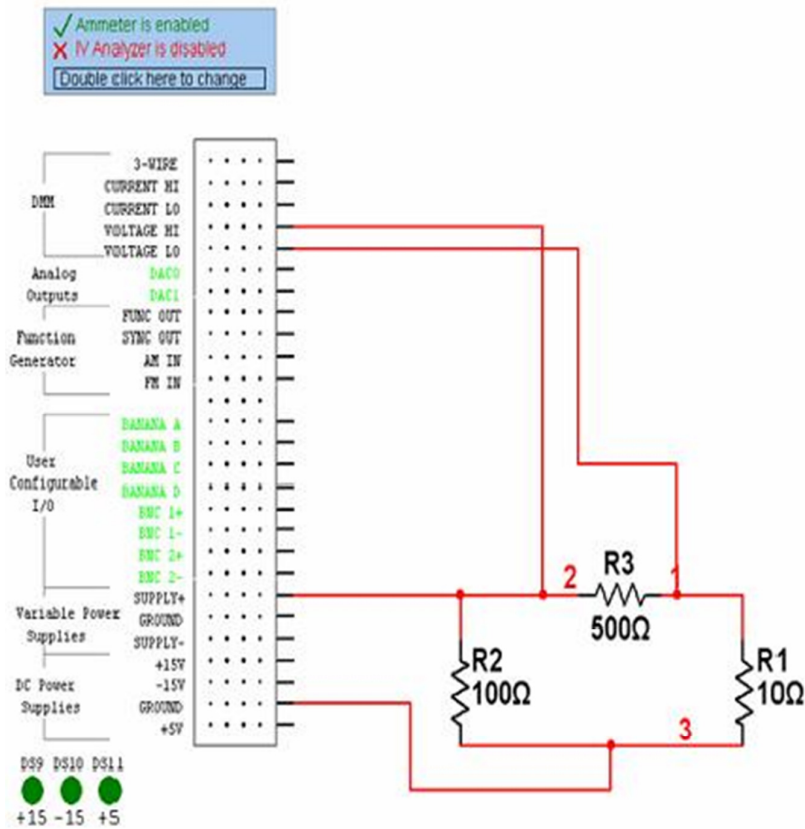


Figure 8

Supply voltage (Volt)	Voltage across R3 (Volt)
1	
3	
5	
7	

Instructions (Part 2):

- Construct the circuit as shown in Figure 9.
- Measure the **current** that flows through the entire circuit for the following VPS voltages:
1V; 3V; 5V; 7 V
- Complete the table below figure 9.

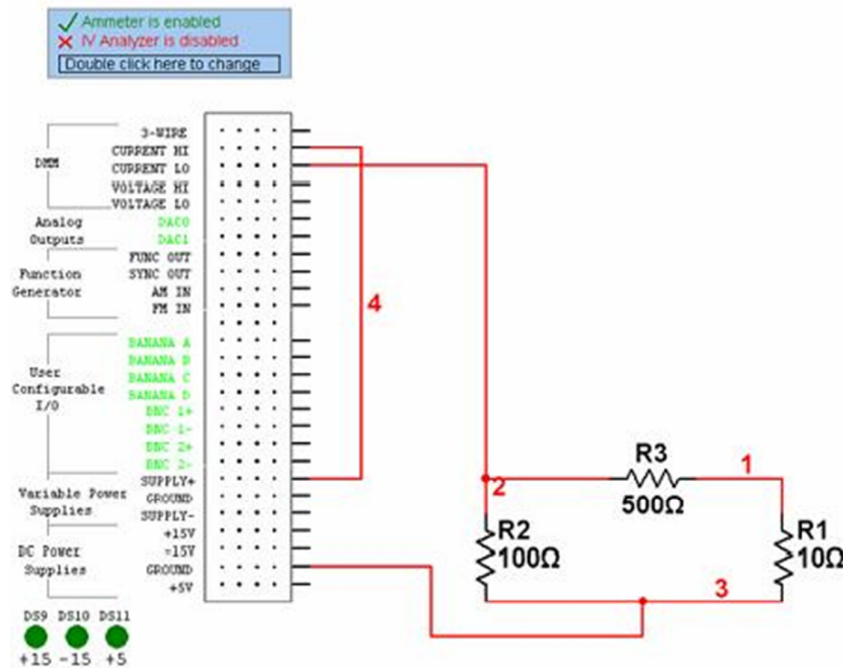


Figure 9

Supply voltage (Volt)	Current through circuit (Ampere)
1	
3	
5	
7	

Now log on to ClickUP and fill in your answers in the:

PRACTICAL 2 ONLINE EVALUATION

Remember to use the decimal point and not the comma for number separation.